Raw Data Storage Structure

RecoMart Recommendation Pipeline | Task 3 Deliverable

1. Overview

Raw data ingested by the RecoMart pipeline is stored in a local data lake using a structured folder layout partitioned by source, type, and timestamp (YYYY-MM-DD). This ensures that every pipeline run produces an isolated, traceable snapshot of its input data.

2. Folder Structure

```
data/
 raw/                              <- Raw ingested data (data lake)
    kaggle/                       <- Source: Kaggle dataset
       product_recommendations/  <- Data type
           YYYY-MM-DD/           <- Date partition
               content_based_recommendation_dataset.csv
    generated/                    <- Source: Simulated REST API
        user_interactions/        <- Data type
            YYYY-MM-DD/           <- Date partition
                user_interactions.csv

 validated/                        <- After validation stage
    validated_products.csv
    validated_interactions.csv

 prepared/                         <- After cleaning & encoding
    prepared_products.csv
    prepared_interactions.csv
    artifacts/
        label_encoders.joblib
        products_scaler.joblib

 features/                         <- Engineered features
    product_features.csv
    product_features.parquet      <- Feast offline store source
    user_features.csv
    user_features.parquet         <- Feast offline store source
    interaction_matrix.csv        <- User-Item matrix

 versioning/                       <- Data lineage & versioning
    versions.json                 <- SHA-256 hash + metadata per file
    lineage.json                  <- Transformation lineage graph

 feast/                            <- Feast feature store
    online_store.db               <- SQLite online store

 warehouse.db                      <- SQLite data warehouse
```

3. Partitioning Strategy

Each raw data file is stored under a 3-level partition hierarchy so that multiple pipeline runs remain isolated and historical data is never overwritten.

Partition Level	Description	Example
Source	Origin system that produced the data	kaggle/, generated/
Type	Category or subject of the dataset	product_recommendations/, user_interactions/
Timestamp	Date the data was ingested (ISO format YYYY-MM-DD)	2026-07-19/

4. Data Sources

Source	Format	Ingestion Method	Rows
Kaggle Ecommerce Dataset	CSV	kagglehub.dataset_download()	1,474 raw / 1,018 clean
Simulated REST API	CSV	Synthetic data generator	5,000 interactions

5. Ingestion Scripts

Script	Purpose
src/ingestion/kaggle_ingestor.py	Downloads Kaggle dataset with retry logic (up to 3 attempts)
src/ingestion/api_ingestor.py	Generates synthetic user interaction data (simulated REST API)
src/ingestion/ingest_runner.py	Unified runner that calls both ingestors sequentially

6. Configuration

All paths are centrally managed in config/pipeline_config.yaml under the paths: section. Every path uses relative addressing so the pipeline runs correctly on any machine regardless of where the repository is cloned.

Config Key	Resolved Path	Stage
paths.raw	data/raw/	Ingestion
paths.validated	data/validated/	Validation
paths.prepared	data/prepared/	Preparation
paths.features	data/features/	Feature Engineering
paths.reports	reports/	Validation / EDA
paths.models	models/	Model Training
paths.logs	logs/	All stages